

DSP48A1 Slice Primitive

Figure 1-2 shows the DSP48A1 slice primitive.

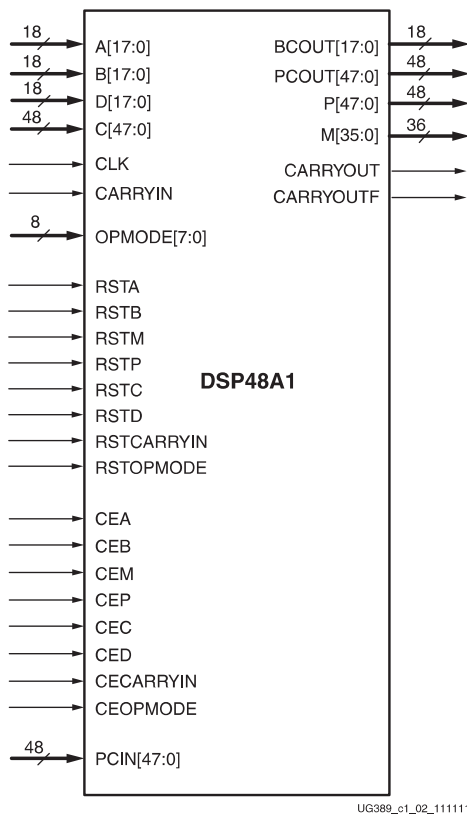


Figure 1-2: DSP48A1 Slice Primitive

Table 1-2 defines the available ports in the DSP48A1 slice primitive.

Table 1-2: DSP48A1 Slice Port Descriptions

Signal Name	Direction	Size	Function
Data Ports			
A	Input	18	18-bit data input to multiplier, and optionally to post-adder/subtractor depending on the value of OPMODE[1:0].
B	Input	18	18-bit data input to pre-adder/subtractor, to multiplier depending on OPMODE[4], or to post-adder/subtractor depending on OPMODE[1:0]. The DSP48A1 component uses this input when cascading BCOUT from an adjacent DSP48A1 slice. The tools then translate BCOUT cascading to the dedicated BCIN input and set the B_INPUT attribute for implementation.
C	Input	48	48-bit data input to post-adder/subtractor.

Table 1-2: DSP48A1 Slice Port Descriptions (Cont'd)

Signal Name	Direction	Size	Function
CARRYOUT	Output	1	Cascade carry out signal from post-adder/subtractor. It can be registered in CARRYOUTREG = 1) or unregistered (CARRYOUTREG = 0). Dedicated cascade that connects only to CARRYIN of adjacent DSP48A1. If not used, leave this output unconnected.
CARRYOUTF	Output	1	Carry out signal from post-adder/subtractor for use in the FPGA logic. It is a copy of the CARRYOUT signal that can be routed to the user logic.
CARRYIN	Input	1	Cascaded, external carry input to the post-adder/subtractor. Should only be connected to the CARRYOUT pin of another DSP48A1 block.
D	Input	18	18-bit data input to pre-adder/subtractor. D[11:0] are concatenated with A and B and optionally sent to post-adder/subtractor depending on the value of OPMODE[1:0].
M	Output	36	36-bit buffered multiplier data output, routable to the FPGA logic. It is either the output of the M register (MREG = 1) or the direct output of the multiplier (MREG = 0). The M and P outputs should not be used at the same time due to potential routing congestion.
P	Output	48	Primary data output from the post-adder/subtractor. It is either the output of the P register (PREG = 1) or the direct output of the post-adder/subtractor (PREG = 0). The M and P outputs should not be used at the same time due to potential routing congestion.
Control Input Ports			
CLK	Input	1	DSP48A1 clock.
OPMODE	Input	8	Control input to select the arithmetic operations of the DSP48A1 slice (see Table 1-7).
Reset/Clock Enable Input Ports			
CEA	Input	1	Clock enable for the A port registers: A0REG = 1 or A1REG = 1. Tie to logic one if not used and A0REG = 1 or A1REG = 1. Tie to logic zero if A0REG = 0 and A1REG = 0.
CEB	Input	1	Clock enable for the B port registers: B0REG = 1 or B1REG = 1. Tie to logic one if not used and B0REG = 1 or B1REG = 1. Tie to logic zero if B0REG = 0 and B1REG = 0.
CEC	Input	1	Clock enable for the C port registers (CREG = 1). Tie to logic one if not used and CREG = 1. Tie to a logic zero if CREG = 0.
CECARRYIN	Input	1	Clock enable for the carry-in register (CYI) and the carry-out register (CYO). Tie to logic one if not used and CARRYINREG = 1 or CARRYOUTREG = 1. Tie to a logic zero if CARRYINREG = 0 and CARRYOUTREG = 0.
CED	Input	1	Clock enable for the D port registers (DREG = 1). Tie to logic one if not used and DREG = 1. Tie to a logic zero if DREG = 0.
CEM	Input	1	Clock enable for the multiplier registers (MREG = 1). Tie to logic one if not used and MREG = 1. Tie to a logic zero if MREG = 0.
CEOPMODE	Input	1	Clock enable for the OPMODE input registers (OPMODEREG = 1). Tie to logic one if not used and OPMODEREG = 1. Tie to a logic zero if OPMODEREG = 0.

Table 1-2: DSP48A1 Slice Port Descriptions (Cont'd)

Signal Name	Direction	Size	Function
CEP	Input	1	Clock enable for the P output port registers (PREG = 1). Tie to logic one if not used and PREG = 1. Tie to a logic zero if PREG = 1.
RSTA	Input	1	Reset for the A port registers: A0REG = 1 or A1REG = 1. Tie to logic zero if not used. This reset is configurable to be synchronous or asynchronous depending on the value of the RSTTYPE attribute.
RSTB	Input	1	Reset for the B port registers: B0REG = 1 or B1REG = 1. Tie to logic zero if not used. This reset is configurable to be synchronous or asynchronous depending on the value of the RSTTYPE attribute.
RSTC	Input	1	Reset for the C input registers (CREG = 1). Tie to logic zero if not used. This reset is configurable to be synchronous or asynchronous depending on the value of the RSTTYPE attribute.
RSTCARRYIN	Input	1	Reset for the carry-in register CYI and the carry-out register CYO. Tie to logic zero if not used. This reset is configurable to be synchronous or asynchronous depending on the value of the RSTTYPE attribute.
RSTD	Input	1	Reset for the D port registers (DREG = 1). Tie to logic zero if not used. This reset is configurable to be synchronous or asynchronous depending on the value of the RSTTYPE attribute.
RSTM	Input	1	Reset for the multiplier registers (MREG = 1). Tie to logic zero if not used. This reset is configurable to be synchronous or asynchronous depending on the value of the RSTTYPE attribute.
RSTOPMODE	Input	1	Reset for the OPMODE registers (OPMODEREG = 1). Tie to logic zero if not used. This reset is configurable to be synchronous or asynchronous depending on the value of the RSTTYPE attribute.
RSTP	Input	1	Reset for the P output registers (PREG = 1). Tie to logic zero if not used. This reset is configurable to be synchronous or asynchronous depending on the value of the RSTTYPE attribute.
Cascade Ports			
BCOUT	Output	18	Cascade output for Port B. If used, connect to the B port of the downstream cascaded DSP48A1. The tools translate the BCOUT cascade to the dedicated BCIN input and set the B_INPUT attribute for implementation. If not used, this port should be left unconnected.
PCIN	Input	48	Cascade input for Port P. If used, connect to the PCOUT of the upstream cascaded DSP48A1. If not used, this port should be tied to all zeros.
PCOUT	Output	48	Cascade output for Port P. If used, connect to the PCIN of the downstream cascaded DSP48A1. If not used, this port should be left unconnected.

OPMODE Pin Descriptions

Table 1-3 shows the OPMODE pin descriptions.

Table 1-3: OPMODE Pin Descriptions

Port Name	Function
OPMODE[1:0]	Specifies the source of the X input to the post-adder/subtractor
	0 – Specifies to place all zeros (disable the post-adder/subtractor and propagate the Z result to P) 1 – Use the multiplier product 2 – Use the P output signal (accumulator) 3 – Use the concatenated D:A:B input signals
OPMODE[3:2]	Specifies the source of the Z input to the post-adder/subtractor
	0 – Specifies to place all zeros (disable the post-adder/subtractor and propagate the multiplier product or other X result to P) 1 – Use the PCIN 2 – Use the P output signal (accumulator) 3 – Use the C port
OPMODE[4]	Specifies the use of the pre-adder/subtractor
	0 – Bypass the pre-adder supplying the data on port B directly to the multiplier 1 – Selects to use the pre-adder adding or subtracting the values on the B and D ports prior to the multiplier
OPMODE[5]	Forces a value on the carry input of the carry-in register (CYI) or direct to the CIN to the post-adder. Only applicable when CARRYINSEL = OPMODE5
OPMODE[6]	Specifies whether the pre-adder/subtractor is an adder or subtracter
	0 – Specifies pre-adder/subtractor to perform an addition operation 1 – Specifies pre-adder/subtractor to perform a subtraction operation (D-B)
OPMODE[7]	Specifies whether the post-adder/subtractor is an adder or subtracter
	0 – Specifies post-adder/subtractor to perform an addition operation 1 – Specifies post-adder/subtractor to perform a subtraction operation (Z-(X+CIN))

DSP48A1 Slice Attributes

The synthesis attributes for the DSP48A1 slice are described in detail throughout this section. With the exception of the RSTTYPE and CARRYINSEL attributes, all other attributes call out pipeline registers in the control and data paths. The value of the attribute sets the number of pipeline registers.

The attribute settings are as follows:

- The A0REG, A1REG, B0REG, and B1REG attributes can take values of 0 or the values define the number of pipeline registers in the A and B input paths. A0REG defaults to 0 (no register). A1REG defaults to 1 (register). B0REG defaults to 0 (no register). B1REG defaults to 1 (register). A0 and B0 are the first stages of the pipelines. A1 and B1 are the second stages of the pipelines. The B1 register is after the pre-adder/subtractor. See the [A, B, C, D, P, and M Port Logic](#) section for more information.
- The CREG, DREG, MREG, and PREG attributes can take a value of 0 or 1. The value defines the number of pipeline registers at the output of the multiplier (MREG)

(shown in [Figure 1-10, page 19](#)) and at the output of the post-adder/subtractor (PREG) (shown in [Figure 1-8, page 18](#)). The PREG attribute controls both the dedicated PCOUT cascade and the P connection to the FPGA logic. The CREG attribute is used to select the pipeline register at the C input (shown in [Figure 1-7, page 18](#)). The DREG attribute is used to select the pipeline register at the D input to the pre-adder/subtractor (shown in [Figure 1-6, page 17](#)). CREG, DREG, MREG, and PREG all default to 1 (registered).

- The CARRYINREG, CARRYOUTREG, and OPMODEREG attributes take a value of 0 if no pipelining register is on these paths, and they take a value of 1 if there is one pipeline register in their path. The CARRYOUTREG attribute controls both the dedicated CARRYOUT cascade and the CARRYOUTF connection to the FPGA logic. CARRYINREG, CARRYOUTREG, and OPMODEREG all default to 1 (registered).
- The CARRYINSEL attribute selects whether the post-adder/subtractor carry-in signal should be sourced from the CARRYIN pin (connected to the CARRYOUT of another DSP48A1 slice) or dynamically controlled from the FPGA logic by the OPMODE[5] input. This attribute can be set to the string CARRYIN or OPMODE5. CARRYINSEL defaults to OPMODE5.
- The B_INPUT attribute defines whether the input to the B port is routed from the parallel input (attribute = DIRECT) or the cascaded input from the previous DSP48A1 slice (attribute = CASCADE). The attribute is only used by place and route tools and is not necessary for the users to set for synthesis. The attribute is determined by the connection to the B port of the DSP48A1 slice. If the B port is connected to the BCOUT of another DSP48A1 slice, then the tools automatically set the attribute to "CASCADE", otherwise it is set to "DIRECT".
- The RSTTYPE attribute selects whether all resets for the DSP48A1 slice should have a synchronous or asynchronous reset capability. This attribute can be set to ASYNC or SYNC. Due to improved timing and circuit stability, it is recommended to always have this set to SYNC unless asynchronous reset is absolutely necessary. RSTTYPE defaults to SYNC. Only reset is directly supported, not set. For this reason, a set (value equals logic 1 upon an applied signal) should not be coded around logic that can be implemented within a DSP48A1 slice.

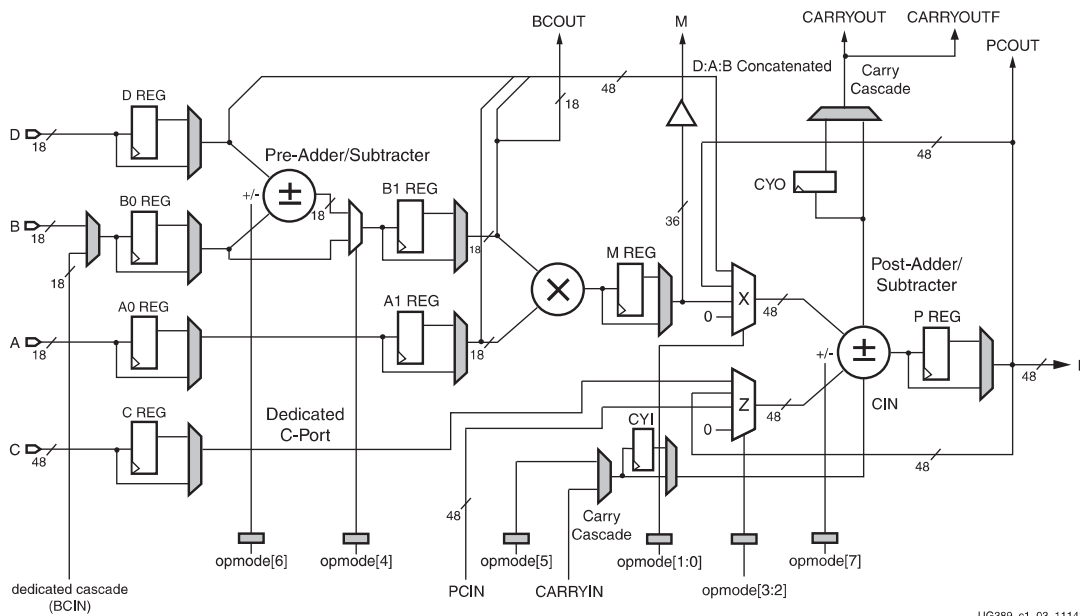
VHDL and Verilog Instantiation Templates

The VHDL and Verilog instantiation templates for the DSP48A1 slice can be found in the ISE tool through the following choice of menus:

```
ISE → Edit → Language Templates → VHDL || Verilog →
Device Primitive Instantiation → Spartan-6 → Arithmetic
Functions → 48-bit Multi-Functional Arithmetic Block (DSP48A1)
```

DSP48A1 Slice in Detail

Figure 1-3 shows a DSP48A1 slice and its associated datapaths. The inputs to the shaded multiplexers are selected by configuration control signals. These attributes are set in the HDL source code or by the User Constraint File (UCF).



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Figure 1-3: DSP48A1 Slice

Notes relevant to Figure 1-3:

1. The 18-bit A, B, and D buses are concatenated in the following order: D[11:0], A[17:0], B[17:0].
2. The X and Z multiplexers are 48-bit designs. Selecting any of the 36-bit inputs provides a 48-bit sign-extended output.
3. The multiplier outputs a 36-bit result that can be optionally registered to the M register.
4. The multiply-accumulate path for P is through the Z multiplexer. The P feedback through the X multiplexer enables accumulation of P cascade when the multiplier is not used.
5. The gray-colored multiplexers are programmed at configuration time. The clear multiplexers are controlled by OPMODE inputs, allowing dynamic changes to functionality.
6. The C register supports multiply-add or wide addition operations.
7. Enabling SUBTRACT implements $Z - (X + \text{CIN})$ at the output of the post-adder/subtractor.
8. B input can be added or subtracted from the D input using the pre-adder/subtractor. Enabling SUBTRACT implements $D - B$ at the output of the pre-adder/subtractor.